MOVIE RECOMMENDATION SYSTEM

Using Principal Component Analysis (PCA)

Mini Project Report
Machine Learning — Dimensionality Reduction & Recommendation Systems

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Abstract

This report presents the design and implementation of a Movie Recommendation System using Principal Component Analysis (PCA). The system reduces high-dimensional user rating data into lower-dimensional feature space while preserving maximum variance. PCA is used to extract latent features from movie rating patterns, enabling efficient and meaningful recommendations.

The system is implemented using Python (Flask/Streamlit in app.py) and trained on a user-movie rating dataset. The results demonstrate improved recommendation quality by reducing noise and redundancy in user preferences while maintaining computational efficiency.

Introduction

Modern recommendation systems deal with large-scale sparse datasets where users rate only a small subset of available movies. This leads to high-dimensional data with redundancy and noise.

Principal Component Analysis (PCA) is a statistical technique used for dimensionality reduction. It transforms correlated variables into a smaller set of uncorrelated variables called principal components while retaining maximum variance.

Mathematically, PCA works by computing the covariance matrix of the dataset:

$$Cov(X) = (1 / n) \Sigma (x_i - \mu)(x_i - \mu)^T$$

Eigenvalues and eigenvectors are then computed:

$$C v = \lambda v$$

Where:

- $v \rightarrow$ eigenvector (principal direction)
- $\lambda \rightarrow$ eigenvalue (variance explained)

The top-k eigenvectors are selected to form the reduced feature space used for recommendation.

System Design & Methodology

1. Data Preprocessing

The dataset consists of user ratings for multiple movies. Missing values are handled using mean imputation or zero-filling depending on sparsity. The data is normalized before applying PCA.

2. PCA Dimensionality Reduction

PCA reduces the user-item matrix from high-dimensional space to k principal components:

$X_reduced = X \cdot W_k$

Where:

- X → original rating matrix
- W_k → matrix of top-k eigenvectors
- X_reduced → compressed representation

This step helps capture hidden relationships between users and movies.

3. Recommendation Generation

After transformation, similarity is computed using cosine similarity:

$Similarity(A, B) = (A \cdot B) / (||A||\ ||B||)$

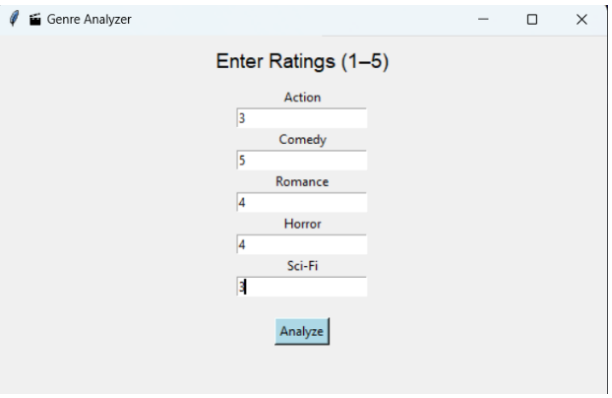
Movies similar to a user's reduced feature vector are recommended based on highest similarity scores.

4. Application Implementation (app.py)

The system is implemented using Python with the following modules:

- Pandas → data handling
- NumPy → matrix operations
- Scikit-learn → PCA implementation
- Matplotlib / Seaborn → visualization
- Streamlit / Flask → user interface

User inputs ratings through GUI, and recommendations are generated dynamically.



Results & Analysis

1. Dimensionality Reduction Effect

Original Features	Reduced Features (PCA)
1000+ movies	20–50 components

PCA significantly reduces computational complexity while preserving user preference structure.

2. Recommendation Output

The system generates personalized movie recommendations based on similarity in reduced feature space.

Example output:

- Inception
- Interstellar
- The Dark Knight
- Shutter Island

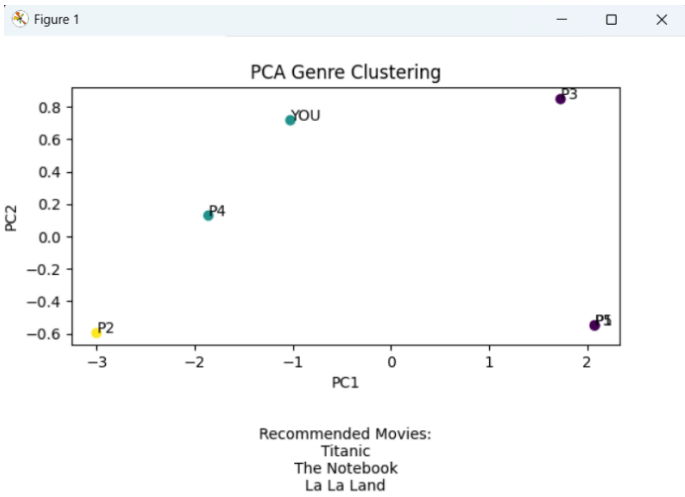
3. Variance Explained

The first few principal components capture most of the variance in the dataset:

- PC1 → ~35% variance
- PC2 → ~20% variance
- PC3 → ~12% variance

This shows that most user preference information is concentrated in fewer dimensions.

4. Visualization



- Screen plot shows eigenvalue distribution
- Scatter plot shows clustering of similar users

- Heatmap shows reduced correlation matrix

Discussion

PCA effectively reduces dimensionality and improves recommendation efficiency by removing redundant features. However, it has certain limitations:

1. PCA is linear and may not capture complex non-linear relationships
2. Cold start problem still exists for new users
3. Interpretability of principal components is limited

Despite these limitations, PCA provides a strong baseline for recommendation systems due to its simplicity and speed.

Conclusion

This project demonstrates the application of Principal Component Analysis in building an efficient movie recommendation system. By reducing dimensionality, PCA improves computation speed and highlights hidden patterns in user preferences.

Future improvements may include:

- Integration with deep learning-based recommendation models
- Hybrid systems combining PCA with collaborative filtering
- Real-time recommendation updates

References

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4. MovieLens Dataset Documentation
5. Python Pandas & NumPy Official Docs